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**Analysing relationships between museums and local economic
activities by mobile phone data in Moscow**

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Abstract

The paper explores the impact of museum on economic development of city neighbourhood. Over the last thirty years there have been discussions that museums can regenerate city economy and help local businesses. However, papers could not provide enough evidence to prove it – authors had to work with aggregated city-wide statistic that could not identify museum's contribution to city economic power without capturing the effect of other initiatives. To fill this gap, this study uses detailed mobile phone app data on pedestrians in Moscow who based on their behaviour is attributed to *rich* and *active youth* categories. The significance of relationships between museums and economic activity was enabled by controlling for such parameters as walkability, street connectivity and transport accessibility. To calculate the strength and direction of local relationships between museums and target pedestrians, OLS regression and geographically-weighted regression (GWR) were built with considering normal, lockdown and recovery times. The results suggest that, in general, museums are not associated with economically active categories but in individual cases such link may exist in residential areas where museums act as cultural centres. To detect which museum characteristics are important for local economic development, further investigations are required.

Key words: culture-led regeneration, museums, GWR, walkability, economic activity, mobile phone data

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1. Introduction

After the success of Guggenheim museum of Bilbao which helped the government to regenerate economic and social environment of Basque country (Guasch and Zulaika, 2005), cultural institutions, especially museums, are viewed as a 'treatment' of urban deprivation and degradation. Policy-makers around the globe have been trying to repeat the success of Bilbao by opening new museums and restoring old ones. Some of them have been expecting higher density of tourists (Koukoulis et al 2019; Lefebvre 1974; Mavragani 2018), whereas others have been foreseeing full regeneration of place that affects not only visitors but also residents (Landry 2006; Guy et al 2002; Clark and Maceer 2008).

Concerning the impact on tourist, most papers agree that museums make a city more popular destination for tourists. To support this, they compare the dynamic of tourist numbers in cities before and after opening the museum. As a result, they observe higher volume of spendings on leisure activities and additional transport costs that generate additional revenue to the city budget. The second assumption about museum power to regenerate cities is supported by much fewer studies but it still is the focus of attention of policy makers that is why is important too. They argue that cultural institutions, including museums, can serve to make city a better place to live, work and interact with community - among expected outcomes are new workplaces, more vivid street life, and higher economic activity (García, 2004; Landry, 2006; Grodach, 2013). Among other benefits are enhanced the reputation of city for investments. Museums are believed to form a new image of cities – more modern and creative- which associates with well-educated and high income people to whom investors trust (Merriman 1991; Walsh 1994). As a results, in the cities with a reputation of cultural places investors more willingly support local businesses and corporations are more likely to open new branches (Bailey, Miles and Stark, 2004; Leary, 2013).

However, the evidence to support these conclusions is weak. Since 90-00 when this idea was the most popular, there has been no study that would assess results using new types of geodata or applying methods that could detect the statistical significance of cultural interventions. All conclusions have been inferred from observation of several cases happened in limited geographic area –Europe and have not had sufficient quantitative support neither of the level of museum contribution in observed changes nor to how

stable and significant these trends are. The risk even increases when we try to apply proposed approaches to assess the impact of a museum in a big city whose image both for tourists and investors consists of multiple factors. In such case a city-wide statistic used by most previous studies would not help but neighbourhood-scale assessment would be necessary. However among all reviewed papers published during the last 30 years only one has found statistically significant local change due to museum (Xavier, 2018), this is an increase of rental prices in museum neighbourhood. Author discovered that the closer to cultural places, the more expensive housing is – which is rather negative for citizens. Apart from it, no other assumptions such as the higher proportion of high income people in residents or active growth of local business have been proved.

At the same time, nowadays the ability of place to attract economically active population is crucial. Having faced the current global situation when travelling activities are limited because of COVID-19 pandemic, cities have to relocate their resources to meet interests of their citizens (Bartik *et al.*, 2020). Without tourists and business travellers they are only source of money for local businesses. Following the marketing reports (Deloitte, 2019; Feedvisor, 2019) about Russian consumers, among them affluent and young groups make up the most attractive audience for street businesses– first generate 60 per cent of revenue of retail, second spend more than other groups in restaurants and cafes. An understanding of museums' influence on them therefore would bring important insights for city government and local businesses.

Based on this observation, this research seeks to find out the level of impact that museums have on wealthy and young citizens walking at their neighbourhoods. To accomplish the goal, the paper uses affinity scores to luxury and youth-oriented brands calculated based on mobile phone app data as a proxy of this measure.

Within the context of increasing availability of data on pedestrians movements due to emerging GPS technologies, the paper focuses the museums impact at neighbourhood scale. This is drawn on the work of Jane Jacobs and some future papers which deemed pedestrian figures strictly related to the quality of built-on urban environment other local factors such as walkability, transport accessibility and street connectivity were added to analysis. To receive them open data from Open Street Map and Moscow open data portal was applied.

Apart from the academic value, this paper may be useful for businesses. As soon as companies know at what extent museums are able to attract their consumers, they can make decisions about opening up a new store or cafe near them.

The objectives of the study are following:

- Provide a review of current approaches to assessing museums impact on the local economic development
- Choose the area which enables relevance of results for different places due to having zones with different economic and social levels
- Measure similarity of places in terms of their walkability and transport accessibility, and locate museums there
- Calculate popularity of places among young and wealthy people
- Define how museums contribute to the popularity of the place among young and wealthy people
- Compare how much these contributions differ in the pre-COVID-19 period (January-March 2020) and post-lockdown period (May-July 2020)
- Find out local features that most likely define the success of the place

Moscow was chosen as study area and this is explained by several reasons. Firstly, it is a large city with more than 400 museums, most of which have been opened in past twenty years. Moreover, they are located all over the city, which let the study to include a full range of economic and social conditions to models. Secondly, Moscow is very conscious about the local impact of urban renovations. Even though the authorities have never treated museums as a part of it, in light of ongoing city scale transformation their possible economic potential should be relevant to them. Thirdly, studies (Bartik *et al.*, 2020; Rice *et al.*, 2020) claim that COVID-19 restrictions set in April-June 2020 in many countries resulted in change of outdoor habits of citizens of big cities who staying cautious about the virus started to prefer neighbourhoods facilities to those in the city centre.

2. Literature review

The purpose of this chapter is to present the development of approaches on assessing the role of museums for city economy and compare them with methodology widely applied to reveal the economic success of place. First, the transformation of views on the role of museums in city over last 100 years will be presented. The chapter will introduce two expected outcomes from museums: bigger tourist flows and more intense economic activity but the main focus will be given to the second expectation. Since the second idea developed in UK, the significant part of this chapter will tell about studies of these countries. Modern scientific and political papers will be critically analysed in terms of data and methods they use, as a result the gap therefore will be identified.

Then we will look at the concept of walkability and based on literature establish the connection between people, street infrastructure and economic activity. In light of these relations, the role of museums for city local economy will be questioned to finally formulate the field of the research.

2.1 Two roles of museum in city transformation

Most of the literature which describe cities renaissance refers to the high significance of culture in this process. Events, cultural facilities, touristic activities, and creative quarters have become the essential parts of development strategies of many declining cities around the globe. Government of different countries, especially in European North and East Asia (Leary, 2013) treat culture as a mean to drive city regeneration. The renovation of city where cultural institute or creative cluster set in the forefront got the name culture-led regeneration. The museum usually takes a central place in this process.

In urban design literature first words about importance culture for city economy were said in 1920s by E. Burgess from Chicago School in the book 'The Growth of the City: An introduction to a Research Project'. There he highlighted the positive difference existed between ordinary and creative quarters of Paris, citing the example of Latin Quartier. Despite being a downtown neighbourhood Latin Quartier is named by author a 'spirit resort' because of its unique atmosphere created by creative communities living there. He highlighted its active social life compared to other Paris districts, thus creating background for further researches.

For the first time, the ability of cultural place to impact city renovation was pointed out by Henri Lefebvre (1974). He noted the high value of places related to local culture for tourism development because they made city outstanding among other destinations. At this time tourism was considered a new source of city income, because tourists willingly spent money on souvenirs, local cuisine and impressions. Therefore, tourism was considered a key of economic stability, and culture, especially recently renovated heritage sites, became a mean to achieve the economic success (Thompson, 2014)

The benefits of making a city touristic place using its cultural facilities were not however limited by visitors spendings. Iconic museums and art-centres located in astonishing modern buildings like centre Pompidou in Paris or Guggenheim museum in Bilbao created a new positive branding of cities which investors desired to fund. The last project became so successful for Basque country revitalisation (Guasch and Zulaika, 2005) that successful results of new initiatives of other cities got name 'Bilbao effect'.

It is noteworthy, that tourists is not the only benefit of museums for cities considered in literature, the effect is usually wider. For example, in case of Bilbao they told about economic regeneration of post-industrial city which created new employment and social opportunities for its residents (Grodach, 2013). More examples where museums impacted residents wellbeing can be found in UK cities such as Liverpool, Newcastle, Reading (Jones and Evans, 2008). There, former industrial sites, stayed in decay in 80s, have been transformed into new work or public spaces, creative clusters and private residences, and in most cases museums were part of projects (Lehmann, 2019). These initiatives created cities of new urban economy (Jones and Evans, 2008; Tallon, 2013) which is characterised by prevalence of consumption over production and aspiration of places to be socially and physically successful.

Drawing on this numerous reports by public bodies such as DCMS (2006), UNESCO (2009) and Heritage Lottery Fund (HLF) (2008) called museums catalysts of urban development. For example, DCMS noticed that improvements of public transport enforced by tourist needs benefited residents as well. HLF extended these findings by assembling results of several social impact case studies. They inferred about crime reduction, increasing involvement of local communities in social life and improved knowledge of culture among locals.

Apart from the social value, economic significance of museums was considered. One of benefits is an active growth of local businesses around museum. (Tuck and Dickinson, 2015) Associated with leisure and creative activities museums bring pleasure, make feel people relaxed and as a result motivate them to buy. Besides, individual impact evaluation papers foresaw inward investments to the area. (House of Commons London 2004; Gunay and Dokmeci 2012) As such, economic value of V&A museum in Dundee opened in 2018 has been positively assessed by company Ekosgen. In the report(Weaver *et al.*, 2019) they have marked a strong contribution of the museum to the wider Scottish economy in terms of new jobs opportunities and GVA generation.

In some cases, cultural revitalisation of city led to ‘enhancement’ of residents. Johnson and Thomas (2001) suggested to consider culture-led regeneration benefits for human capital and art prosperity. In 1990s the group of studies (Merriman 1991; Walsh 1994) linked emerging cultural facilities with an increase of higher-status and well-educated groups in UK cities. They promoted a vision of class division based on cultural background where the highest class consists of people who frequently visit museums. Among other things they argued that ‘heritage tourism’ defines ‘new middle class’ who is ready to spend money to ‘consume’ culture.

To continue this idea Markusen and Schrock (2006) suggested that wealthier individuals are noted to have a higher demand for artistic products. As such, cultural places, including museums are believed to attract richer people, in particular their money.

However not all examples were considered beneficial for local people. It is often argued that emerging tourist flows in city centre displace locals, because their economic activity makes land more expensive (Crewe and Beaverstock 1998; P.Lorente 1996). As a result, property price growth and more expensive stores are located in the centre. This, for example, happened in Glasgow(Grodach, 2013) where due to renovation works included opening Modern Art museum within European city of Culture program city centre, inhabitants had to move to suburbs and luxury brands mostly unavailable for residents took their places. Furthermore, Bianchini & Parkinson (1993) introduced ‘cultural funding dilemmas’ where underfunding dictates authorities to choose between funding one-off cultural or supporting ‘permanent’ activity such as infrastructures. Last, they say that success is reached when museums are part of citywide renovation strategy applied along with economic and social initiatives.

Therefore, in contrast to museums power to attract tourists which is approved by many, its impact on locals wellbeing is still provoking debates in literature and policy papers.

2.2 Gaps in approaches to measuring museum economic contribution

One of the major reasons of disputes is a lack of holistic evaluation framework (McLoughlin et al, 2005) which would be reproducible and allow policy-makers to statistically infer about museum impact. All prior attempts (Bollo 2013; Spero 2013) to create universal approach to measuring failed because of several reasons. First, they relied on large-scale data whereas the impact arguably exists in limited area which is district or neighbourhood. The bigger is study area, the more it has distorting effect on results (HM Treasury, 2013). Studies usually operate museum visitor data when estimate its direct contribution (Ulldemolins, 2014) but this approach neglects the power of museum building to attract people as well as does not give information how active visitors are in nearby facilities.

Furthermore, to assess such frequently discussed in literature indicators as share of people of creative professions and average purchasing power it is proposed to use city-scale scores because it is very difficult to get more detailed data (European Commission, 2017). Meanwhile, it is a close neighbourhood area whose indicators are required be specific about results on people.

Other issue relates to subjectivity of policy-makers in selecting controlled indicators used to assess results. They choose them based on prior expectations, thus considerably limiting the area of analysis (Clark and Maer, 2008; Morrison *et al.*, 2015).

Situation when different renovation initiatives are running simultaneously also worsen the quality of results. Even though a wide range of indicators is considered it is still difficult to say which of changes were caused by cultural facilities. For example, one of the most meaningful efforts to developing the evaluation framework, CASE (CASE, 2010) program of UK DCSM, which suggested to consider market indicators and growing individual's interest to culture suffered from very general view on results. It was impossible to distinguish museum-led outcomes from others occurred in as well as detect for how long they will stay (Robert G. Hollands, 2014). Moreover, according to the paper of What Works Centre for Local Economic Growth (2016), every city differently understands meaning of indicators, that is why they continue to treat culture

impact evaluation problem as they wish. Thus, there is no and results are not locally dependent.

2.3 Summary of issues with assessing museums economic impact

Summing up, on the one hand, the literature review shows us that historically museums along with other cultural institutes were seen as 'catalysts of urban growth'. They are capable of making city a top tourist destination as it happened to Bilbao or revive abandoned former industrial zones as it happened to Liverpool. In both cases studies say about economic and social impact of museums on but numerically these is a limited evidence of these words. The reliable results exist only for tourist statistic.

The variety of methodologies between cities does not allow to aggregate results and identify if they can be relevant to any city. Moreover, even when marginal value for city economy is found, its high level of aggregation does not help to detect which parts of city were impacted and by which of initiatives. These knowledge is important for investors because it shows the capitalisation of their money as well as companies who is planning to open new office in city. Last, no paper detects the period during which effect lasts.

That is why, considering other approaches to measuring economic success of place in terms of is needed.

2.4 Walkability concept

Meanwhile, in wider urban literature a theory about what places are successful in generating economic activity has been being developed since 1960s. According to Jane Jacobs (1961) the diversity of place which includes mixed uses, permeability, variety in the built environment, and high density determines economic potential of city. She argued that diversity or vibrancy of environment attract people to the streets– the idea later developed by Jan Gehl(2006) who added to it the ability of diverse environment to motivate people to interact with each other and be a part of street life. Due to this sidewalk cafes and street retail gives their clients. Later researches shows that transactions is an important component of the vibrancy of area, the higher is its complexity the more successful place is (Montgomery, 1998)

These ideas were essentially important for US urban planning in the end of 20th century (Porter and Shaw, 2009) where active 'urban renewal' processes led to the domination

of wide highways in urban landscape which did not leave much space for walking activities.

Due to the urgent need for walking environment in US, the diversity started to associate with broad opportunities available for pedestrians and finally has been conceptualized as *walkability*. Walkability reflects the degree to which an area encourages walking trips from the property to the destination (Lo, 2009; Adkins *et al.*, 2017). The main advantage of this concept is its measurability (Steiniger, Rojas and Vecchio, 2019).

Urban planners propose different approaches to measuring walkability as well as different criteria of what makes a place walkable. The factors come from specific features of city planning and a purpose of measurement. Some studies (Harvey, 2014; Dunn, Hanson and Seeger, 2018; Natera Orozco *et al.*, 2020) strictly focus on planning functionality such as street network configuration, size of sidewalks and level of mixed use. Their goal is to establish linkage between streetscape design and liveability. Others (Yang, Xu and Long, 2016; Adkins *et al.*, 2017) draw attention to closeness to multimodal transit stations as well as on accessibility of facilities for people with different abilities. Meanwhile, the third group tends to include non-tangible characteristics of places like safety, convenience, or comfort.

2.5 WalkScore

In first group the 'Street Smart WalkScore' methodology (Walk Score, 2011) which evaluates the content of a land use database and elements of street network configuration stands out. Its main advantage is an easy reproduction for every city, that's why real estate from US to Australia to Japan (Koohsari *et al.*, 2018) use this technique and it shows good quality of prediction. This does not demand specific data, on the one hand, and embraces main features of walkability, on the other.

The score is designed to answer how well a passage suits to walking activity in scale from 1 to 100 where 100 means the best conditions. Methodology suggests to consider a distance to closest points of interest (POI) and quality of road connectivity within 1 km. It applies a weighted approach to land-use diversity assessment, which means that different categories and POIs at varying distance from selected place do not equally affect walkability.

As such, restaurants can gain twenty percent of the total score, while bank offices only seven. Moreover, according to WalkScore walkability is affected by up to six close restaurants but only by one park.

To consider distance algorithm suggests using distance decay function - weights are getting less along with an increase of distance to POI- and after third km all amenities are considered not important.

In addition to categories Walk Score also considers landscape aesthetic - it penalizes result if street intersection density is low or if there are many long buildings within walking distance.

However, this methodology was criticized for assessing potential comfort rather than real (Steiniger et al., 2019). Besides, papers pointed at low weight given to recreational areas in comparison with restaurants (Harvey, 2014) and lack of consideration about diversity (Pivo and Fisher, 2011). Also, the WalkScore was accused of neglecting demographic characteristics and such parameters as safety and air quality- there are alternative studies (Credit 2018; Adkins et al. 2017; Steiniger et al 2019) who include it. Nevertheless, though these results better translate people's perception on walkability, the methodology heavily relies on accessibility and accuracy of the data for city planners which is not often the case. Moreover, a big variety of features, most of which explain dependent events, arises a risk of multicollinearity (Steiniger et al., 2019).

All methodologies, therefore, have limitations, thus selection depends on research target and available data.

2.6 Walkability and pedestrians economic activity

Although many benefits of walkable landscape were identified in urban studies within last sixty years such as the increase of social interactions, the change of body mass index or improvement of air quality (Frank *et al.*, 2006), its impact on business prosperity is our primary concern.

In this paper local business prosperity, economic activity or frequency of consumers' transactions are used as synonyms that imply frequency of commercial transactions - buying-selling, exchange, distribution and so on, happened in limited area within city (Tornqvist, 1977).

There are no studies that establish a direct link between walkability and volume of sales in neighbourhood business but there are those who detect changes in some features related to it (Litman 2018; Snieska, Zykiene, and Burksaitiene 2019; Islam and Clarke 2002). One of these is a portrait of pedestrians. According to multiple studies walkable places attract wealthy and well-educated audience - desirable consumers for many businesses - that they explain by several facts. The first reason of having a high share of wealthy people is a high property value (Pivo and Fisher, 2011; Bereitschaft, 2019). Foot accessibility of major types of POIs let residents to avoid using transport regularly thus making rental prices higher.

Moreover, some papers (Florida, 2002), (Drake, 2003) tell about a high share of creative workers in walkable places. They explain it their workplaces close location to walkable places and high importance of aesthetic to them. Such audience also tends to spend more money than average person in cafes and street retail. They prefer a personal contact during purchase in a store to the speed of online sales (Pratt, 2008).

Another discovery of prior studies is a positive correlation of number of well-educated people and walkability degree. Higher level of education in its turn associates with higher GDP per capita - well-educated workforce brings more money to companies and they are ready to pay more to attract talent.(Leinberger and Rodriguez, 2016)As such, companies and cities mutually drive the development of each other, thus making places more economically stable.

Meanwhile, we should remember that there is always a chance to get opposite results - it is possible to have low level of sales in walkable place if other factors not considered in walkability assessment cause it. Firstly, according to Zacharias (2001) such subjective factors as fatigue or personal perception of efforts needed to get to POI may impact demand. Street density characteristics, for example centrality measures or intersections density(Bafna, 2003) only partially control it, overall perception depends on person physiological state . Secondly, it may happen because of external non-measurable features such as specific of local urban planning regulatory (K.Credit, 2018). Another reason may be flaws in company intrinsic processes that does not allow business to grow up at desired rate, even though it is located in walkable place. Therefore, when building model the possibility to observe such outlying results should be taken into consideration.

2.7 Walkability and cultural institutions

For this paper, it is important to note that rare studies consider relationships between cultural institutions and walkability, and no studies that focus on museums.

However, there is wealth of studies which explore relationships between walkability and art-related features. For example it is detected by Mack, Talen, and Koschinsky (2017) that large arts-related businesses that serve wealthy clients are more likely to locate in walkable block groups because they can afford high rental price of these places.

Other papers analyse the linkage between walkability and historical sites that often serve as museum. In such there are papers by Zayed (2016), Abdulla et al (2017) and Abujaber (2009), who put at the core of their researches historical quarters of ancient Middle East cities. Some of them explore the impact of walking facilities on heritage sites which is irrelevant for the current study. Others are closer – they investigate possible changes in walkability explanatory variables which historical environment may cause. But they also are not capable of saying a lot about culture's value to urban development. Their weakness is an over-focus on tourists – they assess the quality of pedestrian networks in heritage districts only to the extent that they attract tourists and completely overlook residents' needs. With this pool of studies, there are no others which explore culture's impact on pedestrians outside historical centres. That is why studies that will consider historical and cultural features when calculating walkability in its general meaning are demanded.

2.8 Measuring pedestrian activity

There are different approaches to collecting knowledge about pedestrians starting from surveys to mobile operators' data to citywide WIFI data to GPS tracking. Whilst the first one suits well when qualitative information is important (Steenbruggen *et al.*, 2013), the last one provides more accurate and broaden view on foot traffic (Maddison and Ni Mhurchu, 2009) due to wider outreach. That is why GPS tracking is often applied to analysing tourist behaviour within an urban destination. For example studies use it to identify their mobility patterns, find explanatory factors of their choice (time, budget, personal motivations, interests, travel group composition, and knowledge of the destination) or model their reaction on tourist infrastructure location. The similar to the current paper goal the paper of Wolf et al. reveals. Its authors used GPS tracking data

to differentiate usage levels in tourist sites, in order to acquire an assessment of tourism-related impacts in natural areas. Therefore, using GPS-data for counting pedestrians in this study seems a more justified and reasonable choice.

However not only data format but source influence the final quality of data. The most accessible way in both financial and technological terms is a use of GPS data retrieved from Location Based Social Network (Li, 2006).LSBN data are generated by millions of people from different countries throughout the world (Hu et al., 2015) and, as the number of social network users grows, so does the amount, quality and usability of data. Moreover, practitioners argue that social media offers different visions on diverse aspects of social, economic and political urban life reflected by the user's interests and activities.

This group of sources includes Twitter, Foursquare or Google Place which have open API that enables to researchers a direct access to their content (Kim, Chae, and Park 2018; Padilla et al. 2018; Sloan, 2017). This opportunity represents a technological advance for urban analysis - the ease with which data collection can be done removes many of the constraints associated with traditional methods such as, collection time, accurate geolocation marks and so on.

But it does not mean that these sources do not have constraints. In opposite, their main weakness is a strong dependence on popularity of social network in studied place and on users tendency to pin-point their location(Personal and Archive, 2019). This problem significantly affects the usability of this data for Russian urbanists because Twitter is the least popular social network in Russia according to recent researches Moscow (BrandAnalytics, 2019) whilst Foursquare is not presented on Russian market at all. As such, Twitter data is not capable to show real numbers of foot traffic in Moscow – dataset will be biased on rare twitter users profiles. Moreover, social media data does not differentiate tourists and resident, whereas the paper focus is strictly on residents. Thereby, these sources do not fully meet requirements of this study and need development before applying

2.9 Research question

Based on presented literature review, there is a potential to discover hidden connections between culture's institutions, for example museums, and local economic activity. Prior attempts to assess it suffered from such issues as lack of geolinked data and non-robust

measuring approaches. Therefore, a fair impact of these organisations is still an open question. New approach that would incorporate new types of geodata and use statistically approved methods is needed to measure it.

The walkability concept was proposed in this chapter as an alternative way to assess local economic activity. In studies it has already been proved that people-oriented environment attracts affluent and well-educated audience, arguably most economically effective. But to build the link between mentioned categories and the prosperity of business, local insights from commercial analytical reports should be analyzed. Moreover, the linkage between walkability with museums has not been well studied yet. Summing up, this creates the motivation for the research done within this paper.

The research question of the paper is to explore how museums impact different groups of pedestrians who is highly likely to generate most revenue to local street businesses. Among pedestrians only residents are interesting to us. We discovered in literature the hypothesis about ability of cultural institutions, including museums, to be an urban catalyst and make places economically successful even without tourist money. This paper seeks to verify this statement.

3. Methods

The goal of the chapter is to introduce a methodology and data which we used in the study to create a new approach to exploring relationships between museums and economically active residents

3.1 Study area

To accomplish the research goal, Moscow city has been chosen. There are several facts that make the city a good place for analysis.

First of all, there are *more than 250 museums*¹ located over all city area, including historical centre and residential districts. Therefore, the diversity of museums features along with urban infrastructure characteristics makes results first, more generalizable and second, more informative in terms of scale of museum impact.

Next, Moscow is especially interesting city for observing the effect of *culture-led regeneration because it has never been discussed* or expected by Russian urban planners. In Soviet Union period culture places served exclusively for purposes of education and valorisation of soviet ideology(Kosenkova, 2014). Their role was to translate cultural identity of the region or nation to all its citizens and rare tourists. In modern Russia the political discourse has shifted to rather economical problems. As a result, museums lost their ideological function but still the majority of them are treated as *cultural custodians* whose role is limited by education function(Kosenkova, 2014). Only some of the most recent museums as Garage museum in Moscow or Yeltsin Centre in Ekaterinburg seek to extend their role in urban life and offer a space for creativity and social interactions. Nevertheless, even these museums have never asserted their regeneration effect.

The similar situation exists in academic papers. Among Russian studies there is no one that assumes an economic impact of museums. Therefore, in Russia when a new museum appears, its location is only explained by land cost and owner desire. As such, Moscow situation let us explore unintended consequences of museum's existence on neighbourhood economy, making research outcomes more valid.

Third reason relates to the *intensive gentrification campaign* the Moscow authorities has been running since 2015(Muravski, 2018). Despite undervaluation of museums

¹ <https://goingrus.com/info/en/russian-cities/siteseeing-in-moscow/museums-in-moscow>

economic potential, they expect other facilities - most recent discussions Fieraru (2018) even embraced creative quarters - to boost city regeneration. To achieve the high level of socio-economic development, they have been improving transport infrastructure and street design, as well as create new open public spaces. New walkable environment is deemed to be able to grow up economic activity at renovated places.

Last reason relates to the *high level of economic activity* in Moscow. Moscow has a vibrant street life, especially in the centre² and, as a result, dense and diverse supply of street services. Knowing, therefore, where a new office will have the highest possible income is crucial for their survival. This became all even more true after lockdown set in Moscow in April-June 2020 period during which 9% of business was closed and others were heavily affected. Urbanists noted that after the lifting of restrictions the extent to which location defines the business success has increased – people started to select place within walk destinations that they deemed safe.

To summarize all listed circumstances, Moscow suits well to the study objectives.

3.2 Data collection

Since Moscow does not collect data by postcode, the smallest available information is a district-level data. However, with the average area of 14 km, taking a district is not very informative for our purposes due to the variety of infrastructure parameters. Instead, hexagons of the Geospatial Indexing System 'H3'(Brodskiy, 2018) were used. Available in resolution of 5 buildings (about 174 m radius) they well suit to describe the effect of museum on next door retail. In this resolution Moscow consists of 8723 cells, by which features values were aggregated.

The data for the research was compiled from a variety of sources:

- Data on POIs and museums location as well as transport accessibility were extracted from Moscow open data portal
- Streets Connectivity data needed to calculate street density and walk time obtained from OpenStreetMap through OSMnx library
- Monthly numbers of different pedestrians groups were provided by Locomizer company based on their mobile application activities.

² here and after the area within Third Transport Circle

3.2.1 POIs and museums data

Data on locations of POIs dated to March 2020 were collected from Moscow open data portal in GeoJSON format from the following datasets:

- Stores
- Restaurants
- Schools
- Universities
- Railway stations
- Metro stations
- Bus stops
- Parks
- Tourist facilities
- Playgrounds

Among museums factors only location was included to the model because it fits with the idea to assess overall impact from museums. Moreover, it is the only factor openly available for all Russian museums - others requires to individually request museums. Location was received from merging 'tourist infrastructure' dataset filtered by museum type of amenity and the data from OpenStreetMap queried directly into Python using Overpass API (tags '*tourism*', key='museum'). It was made because Moscow open data contains only state museums location whereas data on private ones is available on OpenStreetMap. During merging datasets duplicates by name and location were removed to finally get 252 unique museum entities.

3.2.2 Street connectivity data

Apart from POIs categories, input data included street network configuration features. According to urban papers they play role of supplementary drivers of people outdoor activities. To digitize the street network of walk type as multigraph object was collected by using Python library OSMnx which is widely applied in academic world for street scape analysis(Boeing, 2017). In our case it contained 368720 nodes before the removal of deadends deemed negative for pedestrians (Dunn, Hanson and Seeger, 2018) and 354493 after it.

3.2.3 Data on pedestrian activity

To meet study requirements, the GPS data on pedestrians was requested from the company Locomizer. The company provides geolinked recommendations to businesses which they form based on SDK-derived location data. This data comes from mobile applications which track users movements while it is actively used or on background. Due to this information the company links spots, time intervals and mobile users to finally infer about people affinity to some POIs or type of goods that they expresses in score – the higher score, the bigger chance that person will visit the spot .

Such data has several advantages over social media data. First of all, it is more representative - the distribution by age, social class, education level in a sample is closer to real due to the variety of applications (travel, e-commerce, weather, sport and so on) from which it is collected. The second strength is robustness of results. The company remove occasional signals from a sample, so it contains only users with at least twenty signals per month. Last, the Locomizer's approach to data processing mitigates the error of GPS signals. Usually, GPS low accuracy caused by technological limitations or weather conditions (Maddison and Ni Mhurchu, 2009) brings challenges to linking the place and the user. In case of Locomizer, they rely on raw of events rather than one and instead of identifying the one closest POIs, they consider a range of them with different probabilities³.

As soon as Locomizer affinity scores represent user categories, we selected only those that consulting reports about Russian consumers behaviour call the most attractive for street food and retail.

First, according to the report of such companies as Amazon (2019) and Deloitte(2019), millenials (18-32 years) and prime members - people who pay for additional functionality, are the most frequent buyers in all channels - online and offline. For example, in Amazon they make up 56% of all buyers but carry out 90% of sales. Delloite notes that in Russians with above-average income tend to spend more time on shopping than rest (39% vs 24%).

Besides, PWC(2019) underlines an increasing tendency of Russians do regular shopping in nearby stores. This group chooses offline over online but the location of

³ US20150193543A1 - Interest Profile of a User of a Mobile Application - Google Patents 2013

residence matters. Given that biggest diversity of POIs implies higher rental prices, stores can expect consumers with high purchase power.

As for food service, men of age 18-35 are the main audience (75%) of restaurants and bars in Russia and youngs under 25 make up 80% of fast food amateurs(Deloitte, 2019).

Linking these insights with Locomizer categories two scores were chosen for analysis – ‘Active youth’ and ‘Rich men’. Also total number of Residents was added. The specific of Locomizer data enables that only residents data was collected, so no additional processing was needed. The values were received for each hexagon of Moscow area.

To summarize, the received dataset had following characteristics

- Size: 165737 rows
- Day-time interval: 09.00 am – 09.00 pm
- Time period: 01/2019-07/2020
- Periodicity: 1 month
- Locomizer metrics: ‘Residents’, ‘Active youth interest’, ‘Rich men interest’
- Scale: a hexagon, city-scale

Visualizing monthly volumes of residents in Moscow revealed the drastic falls in the periods November-December 2019 and April 2020 (Figure 1) – less than 1 resident in average per hexagon. While April fall is explained by lockdown measures set in Moscow in April-June 2020 during which people moved from the city to countryside, we do not know reasons of November – December fall. Given the short time period available which makes impossible to infer about seasonality of such effect, these three months were deleted from the final dataset to reduce bias.

Besides, hexagons where total number of workers and residents during 16 months was equal to zero were also removed from the dataset. The final dataset consisted of 112032 observations.

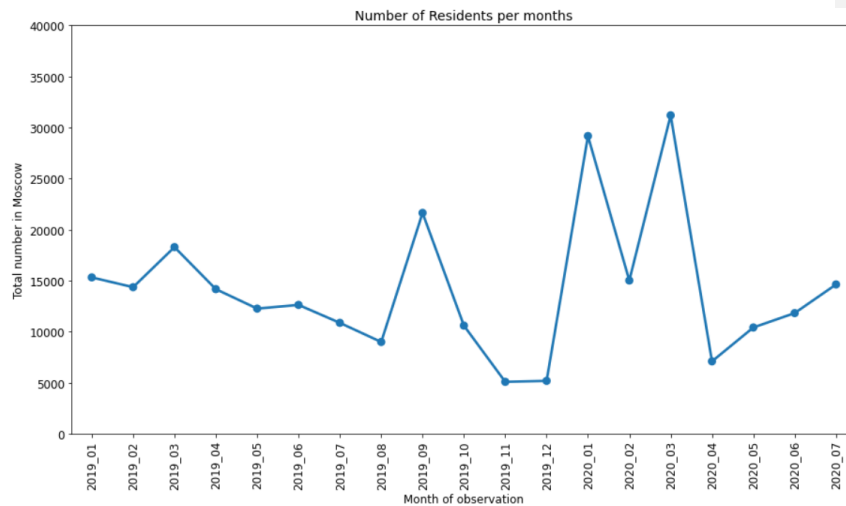


Fig.1 Diagram of *monthly* numbers of Moscow residents in the Locomizer dataset

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3.3 Data processing

3.3.1 WalkScore calculation

To assess walkability, we use WalkScore methodology in this study. Despite the issues, this method is the most adaptable to Moscow realities because it does not require any specific data. In accordance with WalkScore methodology POIs were divided by seven categories (Table 1).

Table 1. Matching WalkScore methodology with Moscow open data

WalkScore category	Dataset name in Moscow open data	Filter	Number of rows
Restaurants	Restaurants	Type of service is either restaurant, or canteen ,or bar	7991
Coffee	Restaurants	Type of service is cafe or buffet	8110
Shopping	Stores	Type of good is not food and type of building is not mall	19135
Grocery	Stores	Type of good is food	19173
Parks	Parks, Tourist infrastructure	Facility is a park, view point or playground	937

Entertainment	Tourist infrastructure	Facility is cinema, theatre, amusement park, or sport	116
Schools	Schools, Universities	All	875

Two notes should be made in relation to how POIs match to suggested categories. First, bank offices category was excluded from calculations in this research, primarily, because there is no such data available for Moscow and, secondly, because with the rapid technological transformation of Russian banks most clients prefer to use online services (Deloitte, 2018), meaning that bank offices do not anymore attract people in Moscow.

Next, the original content of categories has been slightly modified - we added playgrounds to the parks and excluded shopping malls from stores list. First is explained by microdistricts prevalence in Moscow planning which implies a playground for every courtyard, thus making it as popular for family walks as parks.

As for shopping malls, many papers point out the adverse impact of their extremely long grey walls on city aesthetics and as a result on they negatively impact the desire of people to pass by. Therefore, to avoid having different sign of relationship between them and street stores, it was decided to exclude them from analysis.

WalkScore was calculated for each hexagon centroid. To do this, isochrones related to 5, 10, 15, 20 and 30 minutes walk were built using `ego_graph` method from `networkx` library and `convex_hull` method of `OSMnx` library (Fig.2). Then hexagon grid was spatially joined with POIs dataset by every isochrone and amount of every category points was computed. Using isochrones is an alternative to original distance decay function that makes computations faster but saves original order of weights

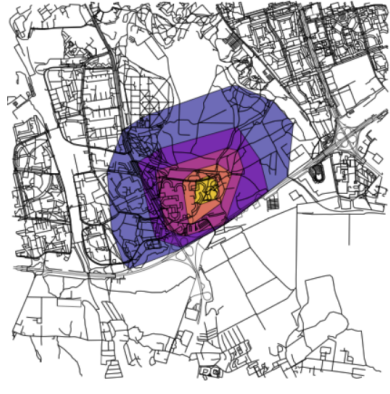


Fig.2 Example of walking isochrones

The final dictionary of points and weights was calculated as following: for every category first n closest POIs were taken, where n represents a recommended number of points, weighted depending on their isochrone and then summarized.

After all scores were ready, they were summed up and normalized by dividing by 13 - total sum of category weights. Then penalty from 0 till 5% was introduced for hexagons where number of intersections within 1 km distance is lower than 200. It therefore gave us a final WalkScore. For analysis purposes it was grouped up into five categories from car-dependent area to walking paradise as the methodology advices.

3.3.2 Calculating connectivity measure

Three connectivity measures were selected for this paper regarding the established in literature their relations with consumers' activity (Taaffe, Gauthier and O'Kelly, 1996; Shashank, 2017) for calculations the area of 1 km radius has been chosen: Intersection density(3), Link – node ratio (4), Connected node ratio (5).

Intersection density is measured as the number of intersections per unit of area which is 1 km in our case. A higher number would indicate more intersections and, presumably, higher connectivity.

$$\text{Intersection density index} = \frac{\text{intersections}}{\text{area}} (3)$$

Link-node ratio employs division of the number of links by the number of nodes, assuming that the fewer streets converge at nodes the lower value of LNR index is.

$$\text{Link – node ratio (LNR)} = \frac{\text{links}}{\text{nodes}} (4)$$

The Connected Node Ratio (CNR) is the number of street intersections divided by the number of nodes (intersections plus deadends). Higher number indicates that there are relatively few deadends and, theoretically, connectivity is higher. (Criterion Planners Engineers, October 2001).

$$\text{Connected node ratio (CNR)} = \frac{\text{intersections}}{\text{nodes}} \quad (5)$$

Sometimes, space syntax centrality measures are encountered. In this paper we did not use them because a preliminary analysis is needed to define bands within which these metrics are most representative (Bafna, 2003).

3.3.3 Variables transformation

Transport accessibility Transport accessibility was calculated by counting number of bus, metro and railway stations within 350 metres that refers to 5 minutes walk – the distance papers call walking.

Museums closeness It was decided to use two variables to capture closeness of museums – their number within 350m and 1 km which relate to 5 minute and 15 minutes walking distance, respectively. Such choice is explained by the desire to look at the effect of close proximity of museum building on pedestrians as well as to assess the total effect of having museum in neighbourhood. To assure they do not explain the same things, multicollinearity level has been accounted.

Land use type. As we discussed above Moscow has areas of two types of land use: commercial and residential. The unavailability of census data in district scale makes find alternative ways to calculate land. That is why it was decided to use Locomizer data on number of workers and residents signals in hexagon to calculate it. Moscow does not house factories nowadays so the area can be roughly divided on commercial and residential area and what is also important districts tend to represent only one of this type. Giving this, the type of area was calculated as following:

$$\text{land_use}_{di} = \text{med}_d \left\{ \text{round} \left(\frac{\text{Residents}_{di}}{\text{Workers}_{di}} \right) \mid 1 \leq i \leq |d| \right\} \quad (6),$$

where d – a district identification, i is a identifier of hexagon belongs to the district
As such, land use equal to one represents residential area and commercial area if zero.

Year season. To identify seasons we used the feature *month* - months affected by corona restricting measures was assigned to separate season. Moreover, being categorical variable requires transformation to numerical. The method *get_dummies* of *pandas*

library in Python was applied to accomplish it, giving us five binary features. One of them was dropped immediately to avoid multicollinearity. Additionally, we added features related to lockdown measures by distinguishing normal time, lockdown and recovery.

Locomizer's affinity scores. To calculate target variables Locomizer scores were normalized by using average score per Moscow. As such, our final variables are *active_youth_interest* score and *rich_men_interest* score in hexagon divided by respective Moscow average.

To account for skewness of *rich_men_interest* variable we used log-transform value because otherwise it can negatively affect the quality of regression.

Calculated values were aggregated per season using average measure.

3.4 Modelling

3.4.1 OLS regression

At first, two OLS regressions were built to test our hypothesis – one per each Locomizer's affinity score. The models allowed to detect the strength and direction of relationships between museums and pedestrians in city scale, while controlling for walkability, street connectivity, transport accessibility, a type of land use, season of the year and situation related to lockdown. Connectivity and transport accessibility measures were added drawn on studies claimed that these features could improve the degree to which walkability measure can explain pedestrians figures (Leyden, 2003; Batty, 2013).

Last, the need to consider the type of area arises when we include museums outside city centre. We assume different associations between people and cultural places in places where social life is rich and where it is poor. It means that even though museums can be found in residential and commercial area of Moscow, their impact on street life will arguably be different.

Besides, previous observations of museums impact stressed out their negative effect on residents life in city centres (Grodach, 2013). Studies argued that they cause a transformation of city centres into uninhabitable places, more touristic than community-oriented. So, introducing additional feature helps to detect if this is true in Moscow.

As for year season variable, it was included to the model in attempt to control for seasonal changes in foot traffic volumes. People are likely to behave in different ways during summer and winter seasons.

Summing up, the common final equation of regressions is as follows:

$$Score = \beta_0 + \beta_1 * X_{museum} + \beta_2 * X_{WS} + \beta_3 * X_{bus} + \beta_4 * X_{metro} + \beta_5 * X_{rail} + \beta_6 * X_{ID} + \beta_7 * X_{CNR} + \beta_8 * X_{LNR} + \beta_9 * X_{season} \quad (1)$$

where *score* – is one of affinity scores, X_i is an independent variable describing feature i ; β_{i0} is intercept parameter; $\beta_{i1..9}$ is a regression coefficient.

Before running the models, we had to test independent variables for multicollinearity issue. To ensure it doesn't exist, VIF measure was calculated and all variables with value higher than 10 (rule of thumb) were eliminated from the model.

When assessing models performance, the focus is on following metrics:

Jackard- Bera - to check assumption of homoscedasticity

Adjusted r-squared – to provide overall strength of relationships

p-value of variable – to assess significance of the variable in model

3.3.2 Index of spatial autocorrelation

The review of urban study papers showed that spatial autocorrelation effect (SAE) often takes place when model assesses building environment (Anselin *et al.*, 1996; Su, Xiao and Zhang, 2012; Yin *et al.*, 2020). Some omitted variables in our model, presumably variables related to district demography and workplaces, can be spatially correlated and cause SAE in the error terms (Case 1991).

To check whether foot traffic flows exhibit spatial patterns, global Moran's I – most common statistic – was computed (Anselin *et al.*, 1996). The computations were made using methods from Pysal library. It evaluates whether the pattern expressed is clustered, dispersed, or random and returns number from -1 to 1 where 0 means random, 100 clustered and -100 – dispersed structure. To define spatial relationships between hexagons a first-order Queen's contiguity spatial weights matrix was built which set 1 if hexagons share a common boundary and zero if no. Statistical significance of the Global Moran's I was obtained through a Monte Carlo simulation (n=9999). This yielded a pseudo p-value, as a threshold in this paper we used 0.05. Positive spatial autocorrelation coefficient reflects the high level of similarity in neighbors which in our

case assumes that people motivate people to walk. In case of negative result, it would mark the tendency to kind of competence – a high crowd density in one place causes low in its neighbor and vice versa. When results are close to zero, no patterns in pedestrian distribution unexplained by OLS are left.

3.3.3 Geographically weighted regression

If spatial patterns in relationships take place fine-scale evaluation is necessary. In this paper for this purpose geographically weighted regressions (GWR) was built. Similar to OLS, it includes additional spatial factor into equation, thus assessing the relationships between target and independent variable in dependence of location. It is not a very popular method in walkability studies, however some authors still use it (Huang, Wu and Barry, 2010; Yin *et al.*, 2020) or consider to use (Brunsdon, Fotheringham and Charlton, 2010) in the future.

The GWR models were of the following form:

$$y_i = \beta_{i0} + \beta_{i1}X_{i,museum} + \beta_{i2}X_{i,WS} + \beta_{i3}X_{i,ibus} + \beta_{i4}X_{i,metro} + \beta_{i5}X_{i,rail} + \beta_{i6}X_{i,ID} + \beta_{i7}X_{i,CNR} + \beta_{i8}X_{i,LNR} + \beta_{i9}X_{i,season} + \beta_{i10}X_{i,situation} + \varepsilon_i \quad (2)$$

where y_i is Locomizer's score at location i ; X_{ik} is k -independent variable at location i ; β_{i0} is intercept parameter at location i ; $\beta_{i1..10}$ is a regression coefficient at location i ; ε_i - regression residual at location i .

Formulating GWR requires to define spatial weighting function that transforms notion of proximity into coefficient estimates and bandwidth which defines size of area using for building local model. As soon as our target distribution is close to normal, a Gaussian model with adaptive spatial kernels using a bisquare function was built. Such choice enables coefficient estimates to vary continuously with distance and at the same time excludes distant points from consideration. To determine the optimal size for the bandwidth, an automated golden section search method with Akaike Information Criteria(AICc) criterion was employed.

4. Analysis and Results

4.1 WalkScore levels and land use in Moscow

The results of calculations of WalkScore (Figure 3) revealed the full range of score levels in Moscow from 0.7 to 95. The infrastructure is predominately rated as ‘very walkable’ and not only in the city center but also in the outskirts. However, large zones in the west and south-east parts are scored ‘Car-dependent’ and ‘Heavily car-dependent’. Empty spaces indicate locations where we did not have Locomizer scores for pedestrians and residents; most of them refer to parks or industrial zones. Overall, the map shows that the further from the city center the more places are rated as car-dependent.

Defining the type of land use based on Locomizer data, we found out that the Moscow centre is fully commercial while outskirts are mostly residential (Fig3). This matches with urbanist notes about then monocentric structure of Moscow. The only exception is Sokol and Aeroport districts in the north-west of Moscow where head offices of big Russian technological companies are located that together employ more than 2000 people.

The correlation matrix shows that links and intersections density are completely correlated those motivating us to remove one of these variables. Given that the literature review identified a higher impact of intersections density on pedestrians, links feature was removed. Moreover, the strong relationship between walkability score and intersections should be kept in mind while building regressions.

Table 2 and 3 present the results of each iteration of building linear regressions for affinity scores *Active youth* and *Rich men*, respectively. First attempts had insignificant variables and high level of multicollinearity that required us to make two iterations for the *active youth* and three iterations for the *rich men* models to achieve robust results. On every iteration variables that had p-value higher than 0.05 were removed. To detect which variables were linearly dependent on each other, we calculated the variance inflation factors (VIFs) and removed the variables that had it higher than 10. Models showed that closeness to bus stations is not important for active young pedestrians, while behaviour of *rich men* does not depend on the number of rail stations around. Also, links-node ration does not impact any of these groups.

Regarding independent variables refer to museums, we found the same trend in both models. Both variables occurred to significantly impact economically active pedestrians but in opposite directions. While museums' presence in 1 km distance is marked as positively important for the popularity of location among studied groups, the area right around museums is rather unattractive for them – the coefficient is below 0. This leads to the conclusion that museums' neighbourhoods are rather attractive for people who consume a lot and museum building adds value to retail and street food in their eyes. However, it is very unlikely that they would attend an exhibition because within a short distance the effect is negative.

Apart from these variables features describing infrastructure and season of the year also revealed interesting patterns. At first, both *active youth* and *rich men* prefer to walk in the commercial centre where outdoor activities and infrastructure are more diverse than in residential areas. Next, in accordance with prior findings, walkability score and intersection density showed a positive connection with pedestrians' intension to spend money on goods – the wider is the diversity of facilities, the wealthier are people this attracts. Also, the importance of distance to metro confirms that metro stays essential for all Moscow citizens regardless of how rich they are. However, other modes of transport are not equally connected with studied groups. As such, bus stations though ignored by youth, seem to be significant for *rich* people. Probably, this happens because on main roads where bus run rent costs are higher, so residents are more affluent than average. Oppositely, railway stations' surroundings seem not pedestrian-friendly; located in the city outskirts, they attract neither *active youth* not *rich* people.

The notable results were obtained in temporal variables. For example, people with *active youth* interests revealed during lockdown higher activity than in normal and recovery time because their share in hexagons increased at that time (the coefficient is positive). As for *rich men*, in opposite, their percentage got less with the introduction of restrictions. Furthermore, recovery time was different from the normal time in terms of young people activity – two related variables were included in the final model - while in terms of rich men activity they showed no significant difference - only lockdown variable mattered.

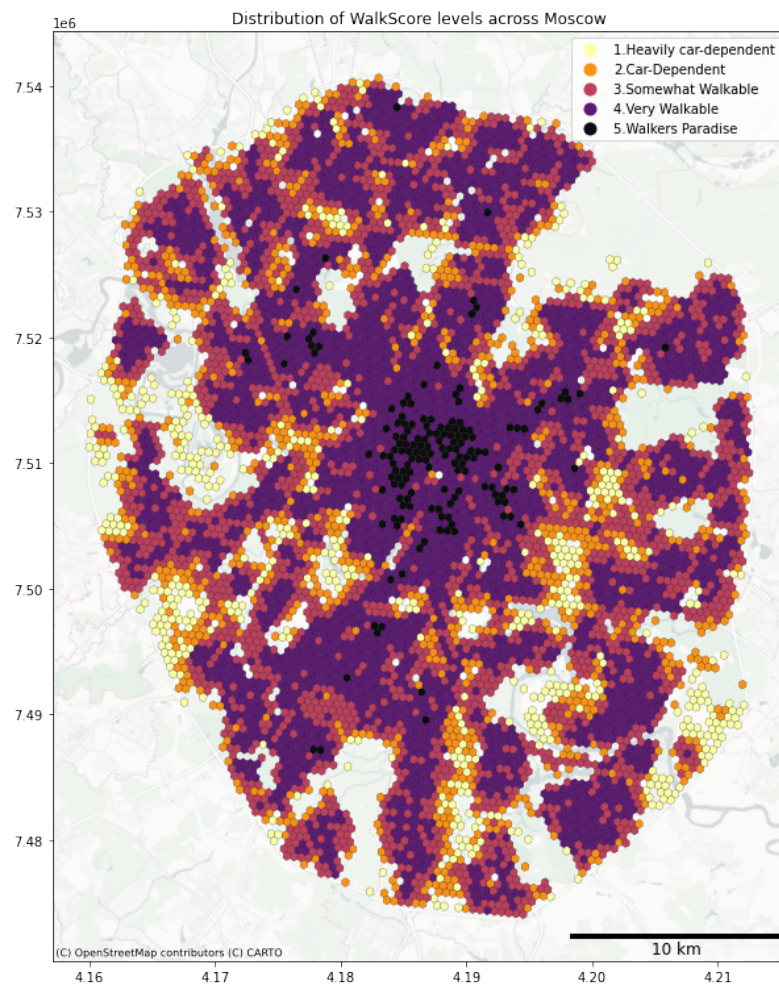


Fig3. Distribution of WalkScore level in Moscow

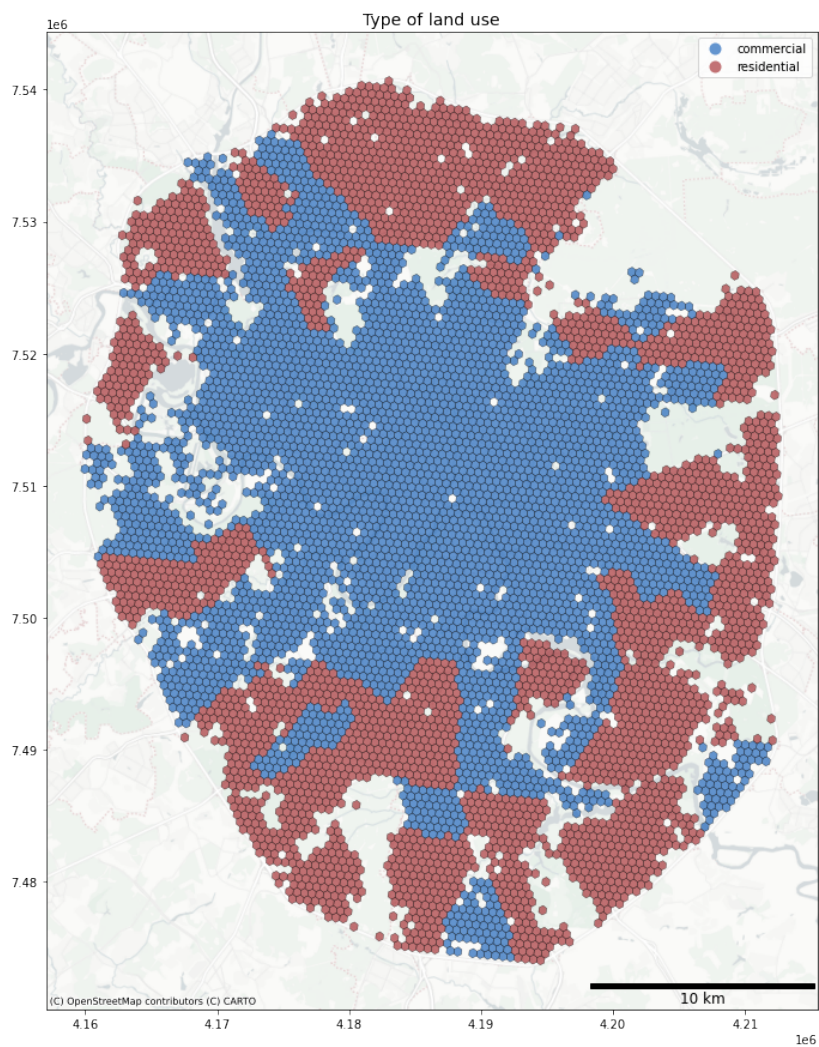


Fig 4. Map of calculated type of land use in Moscow

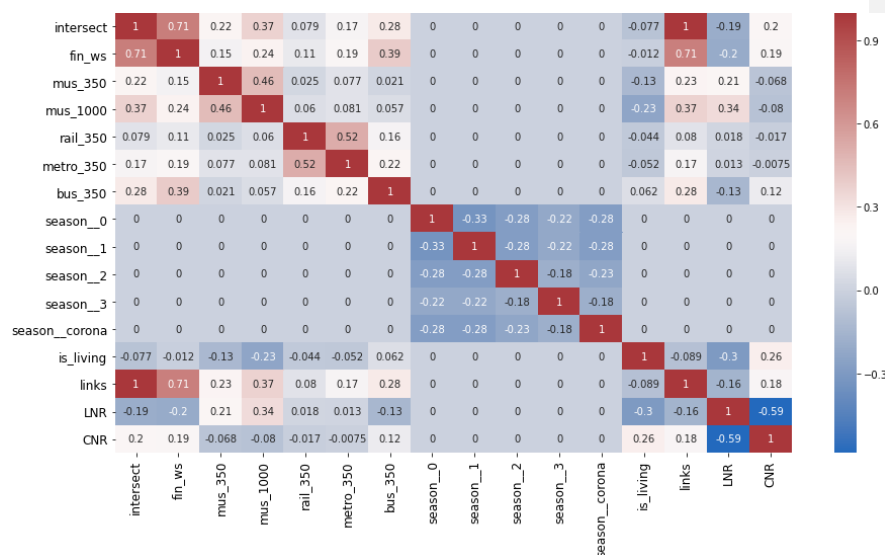


Fig.5 Correlation matrix

Table 2 Linear regression results, Active youth interest

Variable	I iteration			II iteration		
	coef	p-value	vif	coef	p-value	vif
IntDens	0.00002	0.000***	19.963	0.00002	0.000***	4.992
fin_ws	0.0039	0.000***	1.429	0.0039	0.000***	1.217
mus_350	-0.0104	0.004**	1.636	-0.0106	0.003**	1.635
mus_1000	0.0431	0.000***	8.007	0.0445	0.000***	5.596
rail_350	-0.0161	0.000***	1.370	-0.0155	0.000***	1.355
metro_350	0.0796	0.000***	1.438	0.0807	0.000***	1.387
bus_350	0.0007	0.186	1.268			
season_0	0.0401	0.000***	2.250	0.0401	0.000***	2.250
lockdown	0.025	0.000***	3.750	0.0250	0.000***	3.750
normal	-0.0318	0.000***	3.846	-0.0318	0.000***	3.846
season_1	0.0219	0.000***	2.688	0.0219	0.000***	2.688
season_2	0.0101	0.000***	2.688	0.0101	0.000***	2.688
is_living	-0.1518	0.000***	1.105	-0.1516	0.000***	1.102
LNR	0.0016	0.121	28.431			
CNR	-0.2981	0.000***	1.192	-0.2999	0.000***	1.181
intercept	0.8845	0.000***		0.8888	0.000***	
Adjusted R ²				0.312		
AUC				17020		
Jarque-Bera						
Breusch-Pagan						

*** <0.001, **<0.01, *<0.05

Table 3 *Linear regression results, Rich men interest*

Variable	I iteration			II iteration			III iteration		
	coef	p-value	vif	coef	p-value	vif	coef	p-value	vif
IntDens	0.00005	0.000***	19.96	0.00004	0.000***	19.96	0.00004	0.000***	4.99
fin_ws	0.0078	0.000***	1.43	0.0078	0.000***	1.43	0.0079	0.000***	1.36
mus_350	-0.0304	0.000***	1.64	-0.0304	0.000***	1.64	-0.0296	0.001**	1.63
mus_1000	0.0608	0.000***	8.01	0.0608	0.000***	8.01	0.0522	0.000***	5.60
rail_350	-0.0001	0.990	1.37	-	-	-	-	-	-
metro_350	0.1289	0.000***	1.44	0.1288	0.000***	1.12	0.1277	0.000***	1.12
bus_350	0.0065	0.000***	1.27	0.0065	0.000***	1.25	0.0065	0.000***	1.25
season__0	0.0212	0.001**	2.25	0.0212	0.001**	2.25	0.0212	0.001**	2.25
lockdown	-0.0372	0.000***	3.75	-0.0251	0.000***	1.22	-0.0251	0.000***	1.22
normal	-0.0134	0.108	3.85	-	-	-	-	-	-
season__1	0.0544	0.000***	2.69	0.055	0.000***	2.68	0.0550	0.000***	2.68
season__2	0.0442	0.000***	2.69	0.0472	0.000***	2.47	0.0472	0.000***	2.48
is_living	-0.3157	0.000***	1.10	-0.3157	0.000***	1.10	-0.3161	0.000***	1.10
LNR	-0.0101	0.000***	28.43	-0.0101	0.000***	28.43	-	-	-
CNR	-0.6142	0.000***	1.19	-0.6142	0.000***	1.19	-0.6031	0.000***	1.18
intercept	-0.3982	0.000***		-0.4116	0.000***		-0.4405	0.000***	

Adjusted R²

0.21

AIC

209780

Jarque-Bera**Breusch–****Pagan**

*** <0.001, **<0.01, *<0.05

Results of modelling however should be interpreted with caution because some of models indicators detect issues. First, the percentage of explained variance is rather small in both models – adjusted R-squared is equal to 31% and 21%. Next, running Jarque-Bera and Breusch–Pagan tests revealed the strong heteroscedasticity problem. The extremely high value of Jarque-Bera statistic identified non-normally distributed residuals, whereas close to zero p-value of Breusch–Pagan test made us to reject the null hypothesis about constant variance of residuals, thus indicating heteroscedasticity.

Since much variation of target variables left unexplained, it was tested by Global Moran's I to detect if there is a spatial autocorrelation in residuals. This resulted in values 51 and 40 and pseudo p-value< 0.05 which confirmed our assumption and requires to add spatial factor into the models. Mapping them confirms the tendency of residuals to form clusters (Fig 5, 6). It complied with insights from literature about power of people to attract other people to the streets - 'we want to be a part of crowd' Jan Gehl said and our data confirmed this.

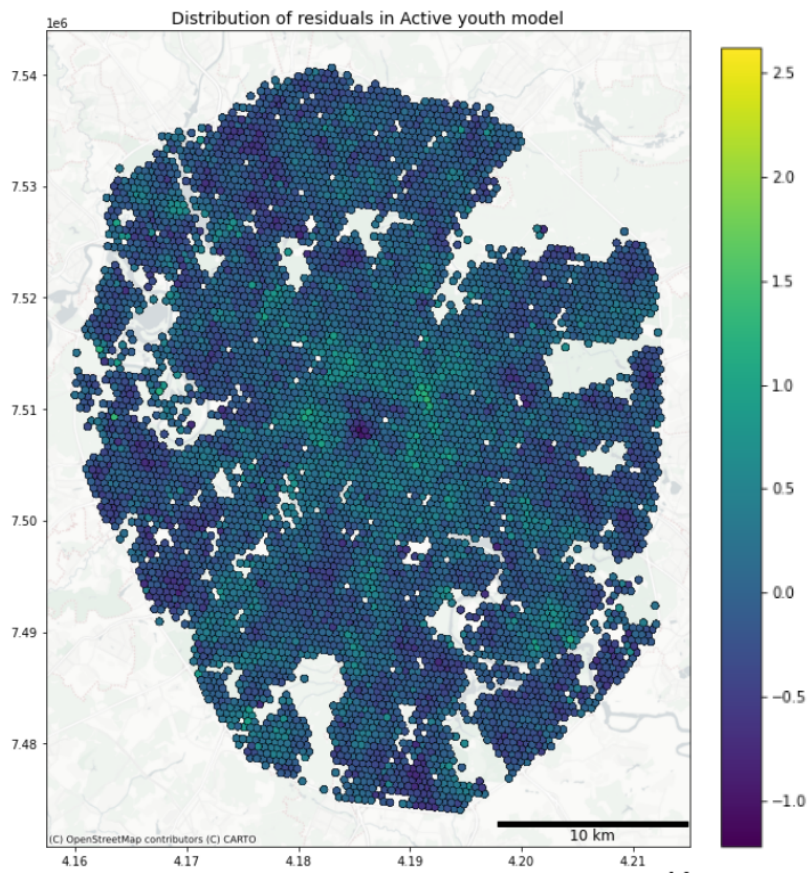


Fig 5. *Residuals of Active youth model*

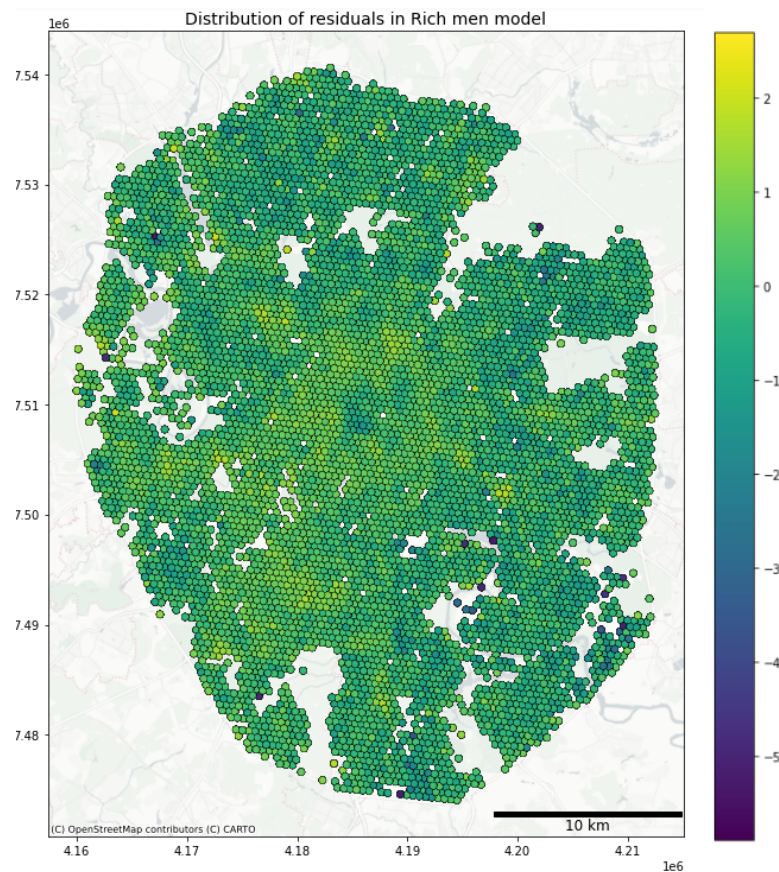


Fig 6. *Residuals of Rich men model*

To incorporate spatial variety geographically weighted regressions were built for two affinity scores using *pysal* library. To run it on the data, we broke it down on three datasets, one per each of time periods: normal, lockdown and recovery. As long as the impact of season on pedestrian behaviour has also been discovered, only period from May 2019 to July 2019 was included to *normal time* dataset. One and half year certainly is not enough to infer seasonality but different relationships of each season with target variables said that we at least should use the months of year for normal and lockdown periods to credibly compare them. To enable one target value per each hexagon, datasets which contained more than one month of observations, affinity scores were aggregate by average. In total six GWR models were built.

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Last, before running the models, optimal bandwidths were found by using golden search from pysal library. The results are presented in Table 4 and 5. Drawing on this, we conclude that rich people's behaviour is explained in bigger scale patterns than young's.

From the model diagnostic, GWR showed the significant increase in the quality of prediction as well as in robustness of results. R-squared improved at least twice in all models, however behaviour of rich men during lockdown in Moscow is still not well explained, probably during this time external factors influenced it stronger than build-on environment. Overall, the result confirms that economic activity for such a big city should be assessed in fine-grained scale rather than city-wide as well as emergency situation made people change their normal lifestyle.

Table 4 *Local regressions results for Active youth*

	Normal		Lockdown		Recovery	
	OLS	GWR	OLS	GWR	OLS	GWR
Adjusted R ²	39%	80%	31%	72%	32%	68%
AIC	-3665	-9527	6643	3092	1710	-2093
Bandwidth	7002	64	7002	85	7002	85

Table 5 *Local regressions results for Rich men*

	Normal		Lockdown		Recovery	
	OLS	GWR	OLS	GWR	OLS	GWR
Adjusted R ²	26%	63%	19%	47%	22%	54%
AIC	9565	5595	17651	15409	13188	10435
Bandwidth	7002	151	7002	154	7002	157

To assess how museums impacted economic active population in different situations, we analysed the significance and direction of local relationships between related variables and each target. In case of GWR they are translated through coefficients – coefficients of museum variables different from zero mean that in this location affinity score is explained by museum presence. Statistic is presented in Table 7 - only hexagons where there is at least one museum within 1 km radius were left to ensure that only museum role will be considered in conclusions. Also, results are aggregated by their proximity to city centre. This follows the assumptions from literature that museums impact on people meaningfully vary by type of land use it is located.

Drawing from the obtained statistic, the hypothesis about general positive influence of museums on economically active citizens that play role of consumers for retail and

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Wang, Z., Fan, C., Zhao, Q., & Myint, S. W. (2020). A Geographically Weighted Regression Approach to Understanding Urbanization Impacts on Urban Warming and Cooling: A Case Study of Las Vegas. *Remote Sensing*, 12(2), 222. <https://doi.org/10.3390/rs12020222>

Commented [QZ4]: You can check one of my recent paper in terms of how to appropriately present GWR results. It's a good finding here that you have a good GWR r2.

Wang, Z., Fan, C., Zhao, Q., & Myint, S. W. (2020). A Geographically Weighted Regression Approach to Understanding Urbanization Impacts on Urban Warming and Cooling: A Case Study of Las Vegas. *Remote Sensing*, 12(2), 222. <https://doi.org/10.3390/rs12020222>

street food has been rejected. With having spatial factor in the models, we didn't not observe a general positive impact of museums on their neighbourhoods. Table 6 shows that in normal times only about 2% of locations around museums in Moscow attract significantly higher percentage of economically active citizens than average.

Moreover, the results confirmed the negative association between the museum location and studied groups of citizens discovered in OLS regressions – in any of the models the number of positive local associations of *mus_350* variable with a target variables did not surpass the number of negative ones. This assumes that expectations of policy makers of direct museum contribution to city economy are unreasonable – its indirect influence on neighbourhoods economic activity should be analysed instead. Due to this, the further analysis will consider only factors referred to 1 km neighbourhood.

Table 6 Statistic of coefficients signs

Model	Independent variable	location	Normal			Lockdown			Recovery		
			-	+	delta	-	+	delta	-	+	delta
Active youth	mus_350	centre	0%	0%	0%	8%	4%	-4%	0%	0%	0%
		residential	4%	1%	-3%	7%	5%	-2%	3%	2%	-1%
	mus_1000	centre	6%	3%	-3%	23%	8%	-15%	5%	9%	4%
		residential	6%	10%	4%	11%	10%	-1%	11%	15%	4%
	mus_350	centre	0%	0%	0%	0%	0%	0%	0%	0%	0%
		residential	0%	0%	0%	6%	2%	-5%	2%	0%	-2%
Rich men	mus_1000	centre	0%	2%	2%	0%	2%	2%	1%	4%	3%
		residential	4%	2%	-2%	8%	3%	-5%	6%	6%	0%

Although the percentage of significant relations is not high, it is noteworthy that in normal times in the city centre museums neighbourhoods (within 1 km from the building) are more likely to attract affluent pedestrians, while outside of it they rather discourage them. However, in 95% of cases there is no correlation at all. Opposite trend has been observed for active youth audience – in 10% of observations museums located in residential areas serve magnets for young citizens though we do not know whether they visit museums or nearby facilities. Meanwhile, their visits to city centre are triggered by reasons different from visiting cultural spots. Youth arguably avoid them –there were only 3% of locations with positive relationships versus 6% with negative ones.

Comparing three situations, we also found out differences in both affinity scores. During lockdown more people deliberately avoided areas around museums. To be

specific, near quarter of central places around museums indicated decrease in youth numbers and near 10% of spots in residential area in rich men numbers. This was due to the government order to go out only for necessity but, due to the higher rental cost in close proximity to museums (Bereitschaft, 2019), essential shops are usually not located there so people visited them less during lockdown.

As for recovery period, in summer months of 2020 the role of museums in citizens life grew – an increase of 10% in non-zero coefficients of museum variables was detected in the both models for *recovery* period. The positive impact on youth people has increased by 5% regardless a type of land use, and by 2% and 4% on rich men in central area and residential, respectively. However, negative impact has also increased mostly in residential areas – by 6% and 1% for *active youth* and *rich men*, respectively. To get an understanding of such changes spatial patterns have been explored.

4.2 Exploring spatial patterns

The insight about how studied groups' behaviour spatially varied over Moscow area as well as how it changed with introduction of restrictions was gained by mapping the sign of local coefficients of '*museum_1000*' variable for every of six models.

The analysis of the maps embodied comparisons of patterns over three time periods and detailed examination of several locations in Moscow with detected positive links for the two affinity scores. Additionally, other features were mapped to find explanations of some behaviour' patterns – they are presented in the appendix 1.

To start, the maps (figures 7 and 8) support our previous findings – coloured spaces vary from one map to another which assume changes in movements of economically active residents over the city due to lockdown measures. Our major concern the sites with homogenous distribution of coefficient sign constitute. As such, in normal times museums in the city centre hardly contribute to variations of studied pedestrians' groups near them – much area is marked not significant except for the small area at the very centre which is highly popular with tourists and as a result not popular with the youth. Meanwhile, at the time when all non-essential businesses were shut down museum places revealed strongly negative association with young people – the much of the city centre is colored blue. Meanwhile, affluent people were not really affected by new circumstances – for them central museums kept to be unimportant.

Recovery period rather repeated normal time patterns as the city centre stopped to be marked negative. The only notable difference is observed in the active youth figures in south-west of the city centre – there in July 2020 green area enlarged indicating a new interest of the youth in some local facilities.

The distribution of walkscore's local coefficients hints at possible reasons of such results. Green color relationships detected at the city centre between walkscore and youth figures imply the constant strong interest of that group in central POIs, no matter how far they are located from museums. It makes sense in Moscow where the centre is densely covered by heritage sights and almost every POI has at least one museum within 1 km distance. At lockdown and recovery time the popularity of pedestrian-friendly places enhanced. However, this does not explain the rich people behaviour. According to maps, they have been staying neutral to the city centre facilities all time. Therefore, more indicators are needed to find out triggers of their behaviour.

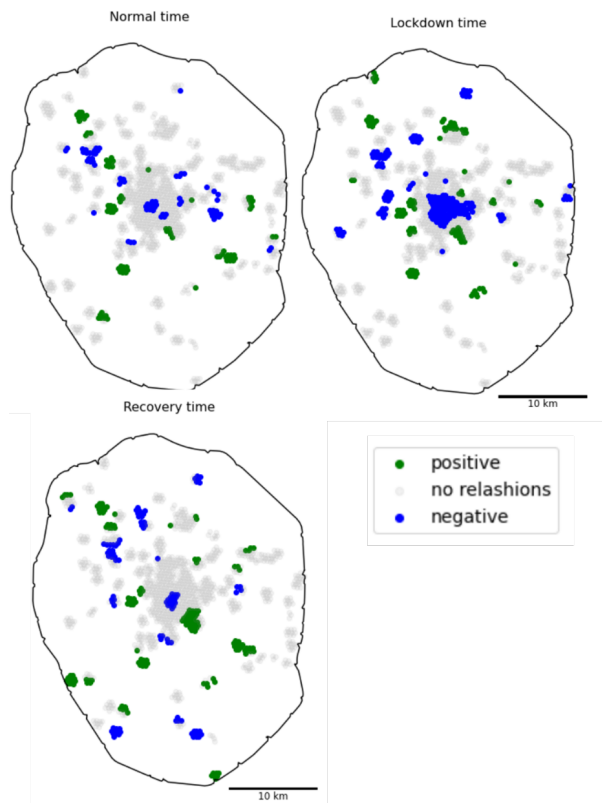


Fig. 7 Distribution of sign of local coefficients of *mus_1000* in Active youth model

Outside of Moscow centre slightly different plot has being developed. First, positive associations between economically active people figures and museums numbers have been identified. At recovery time period their percentage was even higher. On the other hand, negative local relationships were set as well and their number did not get less after lockdown. In active youth behaviour we discovered new deliberate negligence of two places in the south and one in the north that before the pandemic stayed neutral. Rich men also started to avoid more places around museums after lockdown, but at the same time the locations which had attracted them before the lockdown kept and even increased their popular – green areas on the *normal time* map are extended on the *recovery* map.

The fact that at the lockdown time museums stayed closed and was opened again in July was also taken into consideration in further investigations. This let us distinguish

locations where museums attendance is important - at such museum significance was not found in lockdown period but was discovered in the recovery period – from those dependent of neighbourhood development level – there significant link was found on all three maps. When only the first map has significance on some spots, deeper investigation is necessary to detect if behaviour has changed or a spot stayed closed. This is assumed true for both direction of relationships.

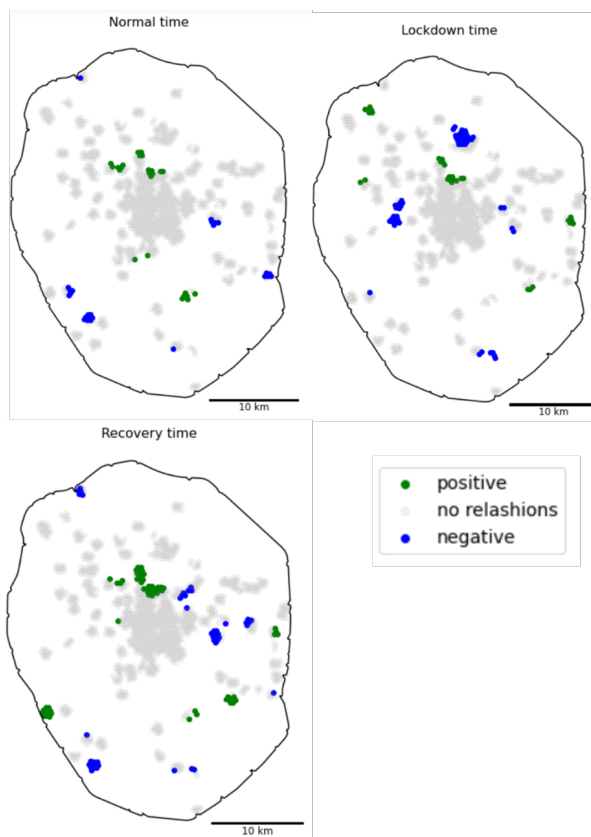


Fig. 8 *Distribution of sign of local coefficients of mus_1000 in Rich men model*

4.3 Insights about young audience behaviour

Individually exploring coloured pieces on *the active youth* maps brought insights some of which contradict with prior expectations. For example, we did not find positive relations for the youth in the Moscow centre. This means that even museums famous

for being loyal to young audience and located in central part of the city, such as Garage and Moscow Museums, failed to bring marginal popularity to their neighbourhoods among the youth – model shows that well developed infrastructure around these sites would anyway encourage young people to visit them.

Seven clusters of green colour referring to positive impact of museums on youth citizens in normal times were found in residential areas outside the city centre. Three of them are located along the Leningradskoe highway in the north-west, two of them exist along Leninskoe avenue in the south-west and two of them are in the east of Moscow. Meantime, five blue clusters stand out on the same map. They locate not too far from the city centre except for one in the north-west.

Seven sites of green colour are areas near Museum of Darwin, Belyaev gallery, Museum *Our origins* and Museum of the Rise of the Machines, Hunting Museum, Museum of CSKA glory, Museum of Russian felt boots and *Bunker-703*. The levels of Active youth score there are 3.7, 2.04, 3.02, 3.86, 4.02, 2.11, 2.9 times higher than average, respectively. Drawing from the logic of defining museum role in significance of the site for the youth above we inferred that Museum of Darwin, Belyaev gallery, Museum *Our origins* and Museum of the Rise of the Machines are popular with the active youth due to their exhibitions while the rest contribute to their location.

Among *negatively affecting the youth* museums there are Museum of nomadic culture. Experimentarium scientific museum, Rohini gallery, Foreign Intelligence Service Museum, Museum of the History of Telephone Communications. However, none of these sites didn't lose its negative impact on the youth in lockdown months which means that it is infrastructure around and not a museum that discourages young people. As soon as an average walkability level is scored *very walkable*, this is an indicator of fragile connection between walkable infrastructure and youth pedestrians.

The information about museums on their websites was analysed in order to find insights which could explain the reason of their success among the Moscow.

Looking through the website information identified that Museum of Darwin⁴ is scientific museum popular for its focus on youth programs and exhibitions. Belyaev gallery⁵ is private gallery which aims at 'being district central of identity'. It is a creative

⁴ <http://www.darwinmuseum.ru/pages/sozdanie-muzeya-osnovateli>

⁵ <https://gallery-belyaev.ru/about>

space that arranges lots of events, including modern art performances, meetings with famous persons as well as educational programs. After pandemic these places kept draw active young audience and even strengthen their influence. Museum *Our origins* and Museum of the Rise of the Machines also serve as local cultural places but with orientation on youth residents of Tekstilschiki district. First one is located in a district youth library building which according to internet reviews is well designed and equipped with a wide range of literature. As such, both services may play role when attract young audience. The second museum is famous for its high level of interactivity. It offers entertaining activities such as laser maze and 3-D printing which are loved by young people and as a result makes museum popular with them.

Examination of features of the museums where the surrounding infrastructure matters discovered that two of them were located inside universities, one existed side by side with a concert hall, two others were at the riverfront area popular for walking. Therefore, the assumption is correct.

Apart from these places higher number of youngsters was discovered around museum of CSKA glory. But this effect is probably confused with the effect of facilities located nearby. The museum is dedicated to the history of famous Russian football club and located in the immediate proximity with the stadium of the CSKA football team. Although sport facilities are considered in walkability score, this is likely not enough to account for crowds visiting football games. Other proof is the fact that during pandemic time when the stadium did not hold games, this site did not reveal significant relations.

4.4 Insights about 'rich men' spatial behaviour

The coefficients of the Rich men interests model mapped on figure 8 show three clusters in the north-west of the city centre which have positive relationships between numbers of affluent citizens and museums. Over May-July 2020 period two of them continued to show significant relationships while one stopped till July when museums opened again. With waiving restrictions, the popularity of whole area increased. This implies the high importance of neighbourhood facilities in general for wealthy audience.

Clusters are located close to each other and refer to the area of Mariina roscha and Savelovskaya districts where three popular Moscow museums and two creative clusters are located. The last may be a reason of uninterrupted popularity of that place during

lockdown because their residents, small business and stores, continued to operate. Moreover, the level of walkability was scored 'very walkable' those indicating the access of local residents to a wide range of facilities which might keep their figures during the lockdown. However, without museums this place would not be so popular with rich people because in terms of other facilities these districts look like any other Moscow residential area which did not show impact on the rich.

In two connected green pieces Jewish museum with Tolerance centre and Tapan centre are located. According to the website⁶, Jewish museum is private museum which hosts a rich collection of artefacts related to Jewish history in Russia. Its case probably is the most close to the examples of European urban regeneration and will be explored in more details later in this paper.

Similar to this one, Tapan centre⁷ also links to one nationality - Armenian. Its plays important role in modern life of Armenian people in Russia and translates its culture through the architecture of its building and the exhibitions. Therefore, both places due to their unique nature and connection with the nationalities historically considered wealthy are able to attract rich persons.

In the second cluster results are more disputable. Even though there is the Esenin Museum, which opened two years ago and is recognised as well-designed and technological, it is more likely that rich people go to the area to visit creative clusters Flakon and Hklebzavod mentioned above but ignored in the models. They bring the atmosphere of chic and creativity the one wealthy residents usually seek. To verify this assumption models advancement is required.

It is noteworthy that in post-lockdown time not only discussed sites kept their popularity but new places in the East of Moscow, such as the museum in Kolomenskoe park, the museum "Datcha" near Izmailovo park and cultural place, became to captivate people with money. A good explanation of this is their closeness to nature - after spending two months at home without having opportunity to move away people started to prefer green spaces to overloaded streets.

⁶ <https://www.jewish-museum.ru/en/about-the-museum/>

⁷ <https://www.vashdosug.ru/msk/exhibition/place/2422858/>

4.5 The case of Jewish museum

A special attention Jewish museum and area around it do deserve. Given the detected popularity of its neighbourhood with affluent people (the affinity score is 2.2 time above average), this museum matches with most expectations expressed by European policy makers and prompts to continue the research. Its history proves that this museum was a part of culture-led regeneration of a former industrial zone.

First, it was opened in Mariina Roscha, an industrial district of Moscow located in north of the centre. During Soviet period it housed factories Borez and Stankolit which produced military equipment. Then factories were closed and the district started to transform to a residential zone of Moscow not different from others. Therefore, the idea to open a large cultural venue 'Garazh' in 2008 and then in 2012 a museum dedicated to Jewish history can be named a start of cultural transformation of the district. Nowadays, it hosts three museums, two theatres and a cultural centre. The cost of living belongs to the first quartile of rental prices in Moscow.

The selected building is also important. It is located in former Bakhmetevsky Bus Garage, the outstanding the monument of Constructivism designed by architectures Konstantin Melnikov and Vladimir Shukhov. Similar to centre Pompidou in Paris or museum of Guggenheim in Bilbao this building dominates surrounding housing, so attract attention of many.

Thirdly, this role is not limited by exhibiting functions. It serves as cultural and educational space for children and adults as well as venue for concerts, lectures and conferences. Also, in the building there are a café, library, souvenir shop, to visit which do not require ticket to an exhibition. They also have outdoor spaces for big events, The museum calls itself 'a multifunctional cultural institution opened for everyone'. Such diversity of facilities creates an environment close to walkable both inside and outside the building - the museum becomes a cultural venue where people are able to relax, interact with each other and surrounding facilities. As a result, following Jan Gehl (Jan, 2006) words, this may motivate them to feel more fluent about committing more economic transaction than usual.

Besides, the museum successfully proved to attract attention of affluent audience. Despite being a private institute, it receives donations from famous people from politics to artists to businessmen, in total bigger than in any other Russian museum.

Therefore, the Jewish museum reveals almost an ideal example of cultural institute that embraced all traits of successful urban catalyst imagined by European policy-makers. However, to understand in which features lead to it, more cases should be reviewed.

5. Discussion

5.1 General economic impact of Moscow museums

Based on the results, we have to decline the hypothesis of museum' general ability to draw the interest of economically active pedestrians to a neighbourhood. In this paper we defined them as *active young* and *rich* people because marketing reports named them the most profitable street food and retail consumers in Russia. Drawn on local-scale analysis of Moscow museums, there is no proof that any museum can generate a higher level of economic activities. Despite found by OLS regression positive relationships between the volume of target audience in a neighbourhood and number of museums, the deeper analysis made us to decline this result. More than 80% of the area with museums did not show any associations with pedestrians' affinity to luxury or youth brands. For businesses this leads to the following conclusion: in big cities like Moscow, they cannot expect to acquire more potential clients when opening a store near museums than in any other place. As such, there is no background for higher economic activity in such places.

Within the context of culture-led regeneration, it seems that there is no reason for local authorities to hope that opening museums will enable additional money coming to the place. In a long-term perspective they are mostly not different from other locations with similar walkability and transport accessibility levels. Therefore, the suggestions of Merriman (1991), Walsh (1994) and Markusen (2006) about the tendency of affluent individuals, who also better educated and more creative, to be near museums were not supported for Moscow. Such places probably motivate wealth tourists to visit a neighbourhood but do not necessarily attract residents with described characteristics. On the other hand, the suggestions of Crewe and Beaverstock (1998) that the only economic impact of museums is an increase of rental prices around was not confirmed too. In this case, we would also observe more *rich* people nearby - walkability related literature showed that higher rents lead to higher-income residents.

Meanwhile, the found exclusions, especially the case of Jewish museum, imply that economic success of museums in a big city is possible when an individual approach to neighbourhood culture-led regeneration is applied and not city-wide, but local characteristics are considered. Residents are not even meant to visit museum every time they are around. When use one of facilities emerged there due to museum, they generate

profit, therefore, confirming the museum indirect economic value. But to enable this, a museum should have an image of the modern, technological and creative space.

But overall, the outcomes confirm worries of researchers who pointed out the flaws of methodologies proposed by local impact evaluation papers and cited in the reports of Heritage Lottery Fund (Bewley and Maer 2014) and UNESCO (2009).

5.2 Strengths of used approach

The methodology applied by this paper to assessing museum impact on economically active residents of Moscow met the majority of methodological weaknesses found in the literature review.

Firstly, access to microscale data on pedestrians affinities significantly extended analytics opportunities comparing to prior efforts. Available GPS-data linked museum impact with neighbourhood characteristics, thus providing deeper and more credible insights than administrative aggregated statistics. The last tries to capture rarely existing city-wide museum value (Montgomery, 1998), especially in such a big city as Moscow. Compared to methodologies of prior studies, this study focuses on a scale where the museum impact is most visible and least affected by other city initiatives. As a result, this allows an understanding of how exactly each museum contributes to the city.

Secondly, the dataset derived from the data on history of pedestrians' behaviour brought the information about citizens' economic potential - a real cause of economic success of the city and motivation for local business to grow. It also simplifies monitoring and evaluation of regeneration for policy-makers as it provides measurable indicators.

Thirdly, the selection of Moscow as a study area enabled representativeness of results. The high diversity of economic and social indicators across the city along with a large number of museums of different size, popularity and status led to generalizable results relevant for cities of different size and type. This helped to make a step toward a holistic evaluation framework discussed in the literature review.

Fourthly, running OLS and geographically-weighted regressions let to distinguish a fair impact of museums and define the importance of spatial features. Adding major infrastructure features such as street connectivity, walkability and transport accessibility to the models let us control for possible external triggers of economic activity and assess only relationships between a distance to museums and the number of pedestrians. Thus,

this approach met concerns of academics about the unaccounted impact of surrounding facilities and gentrification processes on urban economic development.

The analysis of OLS residuals using heterogeneity tests and Global Moran's I revealed the high level of spatial autocorrelation in pedestrian flows. This means that location and neighbourhood indicators are as important for pedestrians as infrastructure in close proximity and should also be considered. That is why in this paper at first time for culture-led regeneration studies spatial features were considered. As a result, the list of features proposed in CASE program (2010) was extended by spatial features unavailable for prior researchers. The received outcomes therefore extended a prior knowledge of features potentially essential for museums economic success in a large city.

5.3 Observing factors of Moscow museums' economic success

Although we cannot generalize the impact of museums on studied categories of residents, the variety of results by locations and by a situation made us believe in the success of museums under particular circumstances. This part brings a list of museums features that the analysis revealed common for positive cases but yet, the number of positive cases is too small to build a model upon them and infer about significant factors. This is a good background to a deeper research.

At first, we observed different trends for central and residential areas. In the city centre where street infrastructure is well developed such places as restaurants, groceries, stores are more important for locals than museums – the relations in the GWR are mostly not significant or negative. Only one spot on the bank of Moscow river with three museums nearby looked attractive that could be explained by the proximity to river, the factor overlooked by walkability formula. General negative tendency became even more evident at the recovery time when most people chose to go only for essentials or to visit the most attractive places. At that time young audience deliberately avoided area around museums – their relationships with museum changed from insignificant to negative while rich people did not provide any reaction on them.

These results match with the prior papers findings about a rather negative effect of cultural initiatives set in the city centre on residents (Grodach, 2013). Quantitative approach together with a diversity of infrastructure characteristics embraced in the models found two possible reasons of it. First of all, the high level of walkability

(Fig.2) in the city centre may be significant. Multiple POIs around, including expensive restaurants and luxury stores are more attractive for the youth than museums. (Deloitte, 2019). The second reason may involve the high density of historic sites in the centre. Since the assumptions in the literature also referred to unique features of museum buildings, they do not stand out in the historic landscape, so do not have a special power to attract people attention. Both suggestions however require verification, for example, by adding to the list of control variables a POI's price segment and buildings traits.

In residential areas the results differ. Even though 90% of the relationships stayed insignificant, zones with significantly higher share of rich and youth audience around museums exist too. Notable change happened at the months of recovery when number of significant museums doubled - four Moscow museums outside the centre were found attractive for youth audience and two for rich people. This highlights museums' role of local cultural centres that they have implemented since travels got limited. In attempt to detect common patterns among successful cases, we found that museums located near natural sides were popular with wealth audience – after the lockdown three museums in Moscow never important for economically active residence revealed the significant increase of that audience near them.

Other reasons were more likely to be linked with museums intrinsic features. As such, five of seven museums were private; all of them regularly hold events and actively promote them in social media. This enables a high attention of young audience. Another essential feature may be an association of a museum with a specific group of people that cases of Jewish museum and Tapan centre established. Yet, the used approach does not allow to detect who caused the economic success of the place: a community or museum. Also, the attraction power of outstanding architecture requires further investigation, two examples is not a sufficient evidence of its value.

5.4 Limitations

Despite the use of new methods and data for the research area , the study has several limitations.

First of all, it is our assumption of economically attractive pedestrians. The choice of categories was explained based on literature and marketing reports insights rather than quantitative assessment of pedestrians' real purchase power. Ideally, knowing the historical number of visited points along with total spending of pedestrians would

improve an estimation of economic potential of the place. Moreover, the artificial categories '*rich men*' and '*active youth*' have an estimation error in representation of real people.

Secondly, despite the thoughtful choice of a study area, it brings limitations to the research. For example, the unavailable demographic data made impossible to consider such features as an employment rate, population diversity and income named important for walkability. As well, rough calculations of land-use type could worsen the results too. Besides, the research within one city, even such a diverse, helps generalize results only in a limited way. For example, an impact of museums in the city centre is not very clear, as Moscow centre is fully historical on,so it doesn't an insight about modern centres.

Thirdly, given that regressions do not establish causal inference, the study cannot detect causality between museums and pedestrians. To do this, pedestrians' data before and after opening museum should be collected. Based on it, a time-series analysis with a use of Granger test could be done to detect causality(Kamarianakis and Prastacos, 2005).

Fourth, the list of studied variables did not include intrinsic museums features. Since the goal was to verify the economic effect of museum's existence such parameters as attendance figures, technological advancement and so on were not added to the models. Using them in further researches would deepen the results and detect which of them are crucial to the economic success.

All these limitations create a motivation for new researches.

6. Conclusion

The historical overview of the role of culture in a city over last 100 years underlined its importance for urban planners and policy makers. Since the growth of urban regeneration activities in Europe, cultural initiatives have been commonly embodied into city redevelopment because of a general belief in their ability to generate money for a city. Such hope was explained by three facts. First of all, cultural places, such as museums and historical sites have proved to be able to attract tourists who willingly spent money on city services. Apart from it, museums were believed to create global branding for cities that attracted big corporations and external investors. This should have led to the increasing number of wealth residents.

This idea came from impact evaluation papers and political reports that explored individual culture-led regeneration cases set in Europe stressing out the significant role of museums in them. The examples of Bilbao, Barcelona and Liverpool where according to these papers ‘one museum’ transformed the whole city economy inspired policy-makers to launch similar programs in their cities with a hope to achieve similar success.

But the real situation was not so simple - concerns about weak methodological approaches applied to assessment of museum contribution have been raised in academic papers. Academics stressed out an overlooked effect of other city initiatives without which an evidence of museums power is not credible. Papers mostly made conclusions on administrative figures changes, meanwhile data to ensure no confusion of a museum’s contribution with other interventions, the neighbourhood-scale analysis has been demanded . But it required fine-grained GPS data, usually unavailable for policy-makers.

Nowadays, when global pandemic spread urges governments to balance in their decisions between people safety and country economic stability, an understanding of real economic value of museums becomes very important. Given the close borders and absence of tourists in cities, it can only be measured in terms of residents money attraction.

Following this, in this paper we proposed a new approach to measuring museums economic value to the city. Based on literature, we concluded that their direct impact is

more evident at local scales and is dependent on economic activity of people who live or pass by there. Therefore, the paper's goal was to explore changes in pedestrians' intention to commit economic transactions triggered by close museum. We measured it by using scores of persons' affinity to goods and services popular with *rich men* and *active youth* in Moscow. To understand the extent to which the distance to museums impacts the score level, factors related to pedestrians' economic activity were taken into account. They included walkability, street connectivity and transport accessibility which we used as control variables in the models. Therefore, we enabled to focus on pure museum impact and mitigate the interference of other local features.

Strength, significance and direction of relationships between museums and pedestrians were found by building geographically-weighted regression. First attempt to use OLS regression failed because of heterogeneity of residuals and the high level of spatial autocorrelation, so we moved to GWR that dealt with these issues due to considering location. This method significantly better explained relationships between museums and pedestrians propensity to consume. As a result, this confirms inability of a city scale statistic to define the role of museums in economic activity of the citizens.

In general, results justified academics' concerns about credibility of evidence provided in policy papers assessing cultural initiatives' impact. Over 80% of Moscow area near museums - this includes places of different land use and walkability level - did not show the impact of museums on local economic activity at all. Using p-value and comparing areas with similar infrastructure revealed that usually museum is not an economic driver – other POIs are more important. But after the release of restrictions both studied groups, revealed a slight growth in positive associations with museums but only in residential area. This indicated the power of museums to be local cultural centres attracting residents at the time when travels on long distances are not safe.

Considering spatial features also helped to found out individual examples of positive associations between museums and economically active residents. All of them were occurred outside the city centre as well as all of them included museums with high level of interactivity and technological developments which gives a background for further investigations. Moreover, exploring Jewish museum we found out its resemblance with famous European examples – avant-garde building constructed by a famous architect, diversity of facilities around, large space thought for social interactions as well as

museum technological advancement repeat European best practices. Therefore, we cannot reject the possibility of a cultural place such as museum to make a city neighbourhood more economically attractive using citizens potential and this study can be a good background for further researches in this field.

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